1 Integral Calculus

1.1 Single Variable Integral Calculus

Theorem 1.1.1: Let $f, g \in \mathcal{R}[a,b]$. Let $D = \{x \in [a,b] : g(x) \neq f(x)\}$. If $\operatorname{cont} D = 0$, then:

$$\int_a^b \mathrm{d}x\, f(x) = \int_a^b \mathrm{d}x\, g(x)$$

Proof,

To prove this theorem, we start with the specific case where a function f doesn't vanish on the set D where $\operatorname{cont} D = 0$.

So, by the definition of content zero, let $\varepsilon \in \mathbb{R}^+$, then we have a finite collection of compact intervals:

$$\{U_i\}_{i \leq n}$$

such that:

$$D \subseteq \left(\bigcup_{i \leq n} U_i\right)^\circ \quad \left|\bigcup_{i \leq n} U_i\right| < \varepsilon$$

The way we argue is the same we argue that if D_f the set of discontinuity of f is content zero, then f is integrable.

Let,

$$U = \bigcup_{i \leq n} U_i$$

Since, f vanishes on $[a,b] \setminus D$, it implies that f vanishes also on

$$V = [a,b] \setminus U^\circ$$

Now, f is integrable over $[a,b]$ since $V \cup U = [a,b]$ and V, U are almost disjoint (i.e., intersection is at most the boundaries) it follows that:

$$\int_a^b \mathrm{d}x\, f(x) = \int_V \mathrm{d}x\, f(x) + \int_U \mathrm{d}x\, f(x)$$

Since, f vanishes on V,

$$\int_a^b \mathrm{d}x\, f(x) = \int_U \mathrm{d}x\, f(x)$$

Since, f has to be bounded on $[a, b]$, this implies that:

$$\left| \int_U \mathrm{d}x\, f(x) \right| \leq (\sup|f[a, b]|)|U|$$

This is actually just the result of the fact that U consist of finite number of compact intervals. However, by definition, of U it follows that:

$$\left| \int_U \mathrm{d}x\, f(x) \right| \leq (\sup|f[a, b]|)\varepsilon$$

As ε is arbitrary, and $\sup|f[a, b]|$ is finite, it implies that:

$$\left| \int_a^b \mathrm{d}x\, f(x) \right| \leq \varepsilon$$

For all $\varepsilon \in \mathbb{R}^+$.

So, this implies that,

$$\int_a^b \mathrm{d}x\, f(x) = 0$$

With this lemma done, the theorem follows since:

$$h = f - g$$

where, f, g follow from the hypothesis.

Here, h is non-zero on a set of content zero. Hence,

$$\int_a^b \mathrm{d}x\, h(x) = 0$$

By linearity of the integral and some algebra, it follows that:

$$\int_a^b \mathrm{d}x\, f(x) = \int_a^b \mathrm{d}x\, g(x)$$

□

Theorem 1.1.2 (The Fundamental Theorem of Calculus):

(a) Let $f \in \mathcal{R}[a,b]$. For $x \in [a,b]$,

$$F(x) := \int_a^x \mathrm{d}t\, f(t)$$

Then F exist and is Lipshitz continuous on $[a,b]$. Moreover, if f is continuous on x, then F is differentiable at x.

(b) Let $F \in C^1([a,b])$. Let $F'(x) = f(x)$. If f is integrable on $[a,b]$, then:

$$\int_a^b \mathrm{d}x\, f(x) = F(b) - F(a)$$

Proof,

(a) Since, f is integrable on $[a,b]$ it follows that f would be integrable over any subinterval within $[a,b]$. Hence,

$$F(x) = \int_a^x \mathrm{d}t\, f(t)$$

exists.

Now, let $\varepsilon \in \mathbb{R}^+$.

So,

$$|F(x) - F(y)| = \left| \int_y^x \mathrm{d}t\, f(t) \right| \leq \int_y^x \mathrm{d}t\, |f(t)|$$

Since, f is bounded,

$$|F(x) - F(y)| \leq \sup|f[a,b]||x-y|$$

Hence, F is Lipschitz continuous.

Assume that f is continuous at x.

So,

$$\lim_{h \to 0} f(x+h) = f(x)$$

Now,

$$\begin{aligned}\left|\frac{F(x+h) - F(x)}{h} - f(x)\right| &= \left|\frac{1}{h}\int_x^{x+h} \mathrm{d}t\, f(t) - \frac{1}{h}\int_x^{x+h} \mathrm{d}t\, f(x)\right| \\ &= \frac{1}{|h|}\left|\int_x^{x+h} \mathrm{d}t\, f(t) - f(x)\right|\end{aligned}$$

Since, f is continuous on $[a, b]$, this implies that it's uniformly continuous on $[a, b]$. So, for all $\varepsilon \in \mathbb{R}^+$, there exists $\delta \in \mathbb{R}^+$, such that:

$$\forall x, y \in [a, b], |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$$

Hence, it follows that if $h < \varepsilon$, then $|x - t| \leq |x - h| < \delta$, hence:

$$\left|\int_x^{x+h} \mathrm{d}t\, f(t) - f(x)\right| \leq \int_x^{x+h} \mathrm{d}t\, |f(t) - f(x)| < \varepsilon|h|$$

Substituting back,

$$\frac{1}{|h|}\left|\int_x^{x+h} \mathrm{d}t\, f(t) - f(x)\right| < \frac{1}{|h|}\varepsilon|h| = \varepsilon$$

Therefore, by the definition of differentiability, it follows that:

$$F'(x) = f(x)$$

(b) Here, we're given that $F \in C^1([a, b])$

Consider,

$$G(a) = \int_a^x \mathrm{d}t\, f(x)$$

Since, f is continuous, from (a), it follows that;

$$H'(x) = F'(x) - G'(x) = f(x) - f(x) = 0$$

vanishes.

So, by MVT, it follows that $H(x) = F(x) - G(x)$ is constant. Now,

$$H(a) = F(a) - G(a) = F(a)$$

Hence,

$$H(b) = F(b) - G(b) = F(a) \implies G(b) = F(b) - F(a)$$

Since,

$$G(b) = \int_a^b \mathrm{d}x\, f(x)$$

we're done.

□

1.1.1 Exercises

Exercise 1. Let (x_k) be a convergent sequence in $\mathbb{R}$. Show that the set $\{x_k\}$ has content zero.

Proof,

Let $\varepsilon \in \mathbb{R}^+$. Since, (x_k) is a convergent sequence, it follows that there exists a K_1 such that:

$$k \geq K_1 \implies x_k \in B_\varepsilon(x)$$

where, $x_k \to x$.

Now, this for the remaining $k-1$ terms, we can cover them using:

$$B_{\frac{\varepsilon}{2^k}}(x_i)$$

balls where $i \in \{1, ..., k-1\}$.

Therefore, we can cover the entire set $\{x_k\}$ with the balls:

$$\left\{B_{\frac{\varepsilon}{2^k}}(x_i)\right\}_i \cup B_\varepsilon(x)$$

The total diameter of which is less than:

$$\frac{(1-k)\varepsilon}{2^k} + \varepsilon = \left(1 + \frac{1-k}{2^k}\right)\varepsilon$$

Since the sequence,

$$\frac{1-k}{2^k} \to 0$$

there exists a K_2 such that:

$$k \geq K_2 \implies \left(1 + \frac{1-k}{2^k}\right)\varepsilon < \frac{1}{2}\varepsilon < \varepsilon$$

Thus, define $K = \max\{K_1, K_2\}$.

Since, $K \geq K_1$, we've that:

$$\left|\left\{B_{\frac{\varepsilon}{2^{K_1}}}(x_i)\right\}_i \cup B_\varepsilon(x)\right| \leq \left(1 + \frac{1-K}{2^K}\right)\varepsilon$$

and, $K \geq K_2$, we've that:

$$\left(1 + \frac{1-K}{2^K}\right)\varepsilon < \varepsilon$$

Thus, we've shown that:

$$\text{cont}(x_k) = 0$$

□

Exercise 2. Let $f \in \mathcal{R}[a,b]$. Suppose that $f(x) \geq 0$ for all x and there exists at least one point $x_0 \in [a,b]$ at which f is continuous $f(x) > 0$. Show that $\int_a^b \mathrm{d}x\, f(x) > 0$

Proof,

Since, $f(x_0) > 0$ and $f(x)$ is continuous at x_0, thus there exists an interval $(x_0 - \delta, x + \delta)$ such that:

$$f(x) > 0$$

Since, we can $f \in \mathcal{R}[a,b]$ it follows we can integrate both sides on $(x_0 - \delta, x_0 + \delta)$. Furthermore, by the order property of the integral, we get:

$$\int_{x_0-\delta}^{x+\delta} \mathrm{d}x\, f(x) > 0$$

Since, $f(x) \geq 0 \implies \int_a^b \mathrm{d}x\, f(x) \geq 0$ by order property of the integral.

By decomposition of domain,

$$\int_a^b \mathrm{d}x\, f(x) = \int_{x\in(x_0-\delta,x+\delta)} \mathrm{d}x\, f(x) + \int_{x\in[a,b]\setminus(x_0-\delta,x+\delta)} \mathrm{d}x\, f(x)$$

By positivity, it follows that:

$$\int_{x\in[a,b]\setminus(x_0-\delta,x+\delta)} \mathrm{d}x\, f(x) \geq 0$$

However, since the remaining integral is strictly positive, it follows that:

$$\int_a^b \mathrm{d}x\, f(x) > 0$$

□

Exercise 3. Let f be an integrable function on $[a, b]$. Prove the following formulas directly from the definitions:

(a) (Scale Property) For any $c \in \mathbb{R}+$,

$$\int_a^b \mathrm{d}x\, f(x) = c\int_{\frac{a}{c}}^{\frac{b}{c}} \mathrm{d}x\, f(cx)$$

(b) (Reflection Property)

$$\int_a^b \mathrm{d}x\, f(x) = \int_{-b}^{-a} \mathrm{d}x\, f(-x)$$

(c) (Translation Property) For any $c \in \mathbb{R}$,

$$\int_a^b \mathrm{d}x\, f(x) = \int_{a-c}^{b-c} \mathrm{d}x\, f(x+c)$$

Proof,

(a) Since,

$$\sup_{[a,b]} f(x) = \sup_{[\frac{a}{c},\frac{b}{c}]} f(cx)$$

This follows because $f([a,b]) = f\big(c\big[\frac{a}{c},\frac{b}{c}\big]\big)$ and $c \notin \mathbb{R}^-$.

It's not an overtly big leap to conclude that given a partition $P \in [a,b]$, we can consider the partition:

$$c^{-1}P = \left\{ \frac{x}{c} : x \in P \right\}$$

Note that $c \neq 0$ is needed here. Using this partition, we have:

$$M_i^P f = M_i^{c^{-1}P} f \quad m_i^P f = m_i^{c^{-1}P} f$$

However, the issue is that $(\Delta x)_i^P \neq (\Delta x)_i^{c^{-1}P}$.

But, from definition, it follows that:

$$(\Delta x)_i^{c^{-1}P} = \frac{x_i}{c} - \frac{x_{i-1}}{c} = c^{-1}(\Delta x)_i^P$$

Therefore, we get the following relationships:

$$\begin{aligned} U_{c^{-1}P} f(cx) &= \sum (\Delta x)_i^{c^{-1}P} M_i^{c^{-1}P} f(cx) \\ &= \frac{1}{c} \sum (\Delta x)_i^P M_i^P f(x) \\ &= \frac{1}{c} U_P f(x) \end{aligned}$$

As f is integrable on $[a,b]$ it follows there exists a sequence of partitions (P_n) such that:

$$\lim_{n\to\infty} \left(U_{P_n} - L_{P_n} \right) f(x) = 0$$

So, consider the sequence of partitions $(c^{-1}P_n)$ of $\left[\frac{a}{c}, \frac{b}{c}\right]$.

$$\lim_{n\to\infty} (U_{c^{-1}P} - L_{c^{-1}P}) f(cx) = \lim_{n\to\infty} c^{-1} \left(U_p - L_p \right) f(x) = 0$$

Thus, $f(cx)$ is integrable over $\left[\frac{a}{c}, \frac{b}{c}\right]$.

Again, using the same relationship, we get that:

$$\inf U_{c^{-1}P} f(cx) = \inf \frac{1}{c} U_P f(x) = \frac{1}{c} \inf U_p f(x)$$

where to take out c and preserve inf we needed non-negativity of c and invertibility of the transformation $c \to \frac{1}{c}$ (requires $c \neq 0$).

(You can also limits with P_n (directly or squeeze))

By the definition of Riemann integrability, it follows:

$$\int_{\frac{a}{c}}^{\frac{b}{c}} \mathrm{d}x\, f(cx) = \frac{1}{c}\int_{a}^{b} \mathrm{d}x\, f(x) \iff \int_{a}^{b} \mathrm{d}x\, f(x) = c\int_{\frac{a}{c}}^{\frac{b}{c}} \mathrm{d}x\, f(cx)$$

(b) Since, $a < b$, it follows that $-b, -a$.

Now, it shouldn't be too difficult to see that;

$$f(-[-b,-a]) = f([a,b])$$

Proving this fact is a simple two-way inclusion argument. Really, the important part is to show that:

$$-[-b,-a] = [a,b]$$

Regardless, the equality of these two sets leads us to the fact that:

$$\sup f(-[-b,-a]) = \sup f([a,b])$$

Thus, this implies that:

$$\sup_{[-b,-a]} f(-x) = \sup_{[a,b]} f(x)$$

Consider P be a partition of $[a,b]$. It follows that:

$$P = \{x_0 < x_1 < \ldots < x_n\}$$

By order property,

$$-P = \{-x_n < -x_{n-1} < \ldots < -x_0\}$$

Instead of using the usual order of summation on partitions, we sum $-P$ by the reverse order. This doesn't affect the sums since we're considering a finite sum.

Thus, in this world we've:

$$\sup_{[-x_i,-x_{i-1}]} f(-x)(-x_{i-1} - (-x_i)) = \sup_{[x_{i-1},x_i]} f(x)(x_i - x_{i-1})$$

Since the terms of sums are equal for arbitrary i, it follows that:

$$\sum_{i=1}^{n} \sup_{[-x_i,-x_{i-1}]} f(-x)(-x_{i-1} - (-x_i)) = \sum_{i=1}^{n} \sup_{[x_{i-1},x_i]} f(x)(x_i - x_{i-1})$$

$$U_{-P} f(-x) = U_P f(x)$$

Again, I draw to note that we're summing $U_{-P}f(-x)$ in the reverse of the order we usually sum up an upper Riemann sum.

Similarly, we also have:

$$L_{-P}f(-x) = L_P f(x)$$

Now, integrability of $f(-x)$ follows from the fact that:

$$(U_{-P} - L_{-P})f(-x) = (U_P - L_P)f(x)$$

So, using the integrability of f, we can draw (P_n) such that:

$$\lim_{n\to\infty}\left(U_{P_n} - L_{P_n}\right)f(x) = 0$$

Thus,

$$\lim_{n\to\infty}\left(U_{-P_n} - L_{-P_n}\right)f(-x) = 0$$

Hence, $f(-x)$ is integrable on $[-b, -a]$.

Again, integrability also gives us:

$$\lim_{n\to\infty} U_{P_n}f(x) = \int_a^b \mathrm{d}x\, f(x) = \lim_{n\to\infty} U_{-P_n}f(-x) = \int_{-b}^{-a} \mathrm{d}x\, f(-x)$$

(3) The argument of (3) is basically the same as (1), (2). The key points are we need to show that sup / inf remain constant over some transformed domain for the transformed function. Finally, spam the property of the integral and them integrability condition.

□

1.2 Integration in Higher Dimension

Definition 1.2.1 (Cartesian Product): Let A, B be sets. Then the Cartesian product of A, B is:

$$A \times B := \{(a, b) : a \in A, b \in B\}$$

□

1.2.1 Rectangles

Definition 1.2.1.1 (Compact Rectangle): Let $R \subseteq \mathbb{R}^n$ be a compact rectangle. Then,

$$R = \prod_{i=1}^{n} [a_i, b_i]$$

where, $[a_i, b_i] \subseteq \mathbb{R}$ is a compact interval.

□

1.2.2 Partitions and Subrectangles

Definition 1.2.2.1 (Partitions of Compact Rectangles): Let $R \subseteq \mathbb{R}^n$ be a rectangle, then a partition P on R is the set:

$$P = (P_1, ..., P_n) = (P_i)_{1 \leq i \leq n} = (P_i)$$

where, P_i is a partition of $[a_i, b_i]$, where, of course,

$$R = \prod_{i=1}^{n} [a_i, b_i]$$

Definition 1.2.2.2 (Sub-Rectangles from Partitions): Let $P \subseteq \mathbb{R}^n$ be a partition of the compact rectangle $R \in \mathbb{R}^n$. If

$$P = (P_1, ..., P_n) \quad R = \prod_{i=1}^{n} [a_i, b_i]$$

Then, given the multi-index:

$$J = (j_1, j_2, ..., j_n)$$

where, $x_{j_i}^i \in P_i$, then we define the sub-rectangle:

$$R_J = \prod_{i=1}^{n} \left[x_{j_i-1}^i, x_{j_i}^i\right]$$

□

1.2.3 Upper and Lower Riemann Sums

Definition 1.2.3.1 (Riemann Sups and Infs): Let $f : \mathbb{R}^n \to \mathbb{R}$ be a bounded function defined on the compact rectangle $R \subseteq \mathbb{R}^n$. Suppose P is a partition of R. Then, for a particular multi-index J from P, we get:

$$M_J^P f = \sup\{f(x) : x \in R_J\} \quad m_J^P f = \inf\{f(x) : x \in R_J\}$$

where sup, inf both exists due to boundedness.

Definition 1.2.3.2 (Volume Element): Let $f : \mathbb{R}^n \to \mathbb{R}$ be a bounded function defined on the compact rectangle $R \subseteq \mathbb{R}^n$. Suppose P is a partition of R. Then, for a particular multi-index J from P, we get the volume element:

$$(\Delta x)_J = |R_J| = \prod_{i=1}^{n} \left(x_{j_i}^i - x_{j_i-1}^i\right)$$

With these both definitions in hand, we can now define the lower and upper sums.

Definition 1.2.3.3 (Riemann Sum): Let $f : \mathbb{R}^n \to \mathbb{R}$ be a bounded function defined on the compact rectangle $R \subseteq \mathbb{R}^n$. Suppose P is a partition of R. Let J be a multi-index from P. Then, we define the upper and lower Riemann sum as:

$$U_P f(x) = \sum_{J \in P} (\Delta x)_J M_J^P f(x) \quad L_P f(x) = \sum_{J \in P} (\Delta x)_J m_J^P f(x)$$

□

1.2.4 Riemann Integral on $\mathbb{R}^n$

Like with the single variable case, we first define lower and upper integrals.

Definition 1.2.4.1 (Lower and Upper Integral): Let $f : \mathbb{R}^n \to \mathbb{R}$ be a bounded function defined on the compact rectangle $R \subseteq \mathbb{R}^n$. Suppose P is a partition of R.

$$Uf(x) = \inf_P U_P f(x) \quad Lf(x) = \sup_P L_P f(x)$$

Finally, we have the definition of the Riemann integral.

Definition 1.2.4.2 (Riemann Integral): Let $f : \mathbb{R}^n \to \mathbb{R}$ be a bounded function defined on the compact rectangle $R \subseteq \mathbb{R}^n$. Suppose P is a partition of R. Then, f is said to be Riemann integrable on R if:

$$Lf(x) = Uf(x)$$

We notate the integral as:

$$\int_R \mathrm{d}x\, f(x)$$

□

1.2.5 Riemann Integral on $\mathbb{R}^n$ on General Regions

Definition 1.2.5.1 (Indicator Function): Let A be a set, then we define the indicator function:

$$\chi_A(x) = \begin{cases} 1 & x \in A \\ 0 & \text{otherwise} \end{cases}$$

By using indicator function, we can extend the definition of f that might be defined in specific domains to rectangles. Namely

Definition 1.2.5.2 (Integration over General Domains): Let $f : D \to \mathbb{R}$ be defined on some $D \subseteq \mathbb{R}^n$ which is bounded. Then, there exists a rectangle such that $D \subseteq R$, such that:

$$g(x) = f_{\chi_D}(x) = \begin{cases} f(x) & x \in D \\ 0 & x \in R \setminus D \end{cases}$$

Then, f is integrable over R if f_{χ_D} is integrable over R. And,

$$\int_R \mathrm{d}x\, f(x) = \int_R \mathrm{d}x\, f_{\chi_D}(x)$$

□

1.2.6 Criterion for Integrability

Definition 1.2.6.1 (Oscillation): Let $f : \mathbb{R}^n \to \mathbb{R}$ be a bounded function defined over a compact rectangle R. Let P be a partition of R. Let J be a multi-index from P. Then,

$$\omega_J^P f(x) = \sup_{x,y \in R_J} |f(x) - f(y)| = M_J^P f(x) - m_J^P f(x)$$

The equality follows from the theorem:

Theorem 1.2.6.1: Let $f : A \to \mathbb{R}$ be a function such that $f(A)$ is bounded. Then,

$$\sup_{x,y \in A} |f(x) - f(y)| = \sup f(A) - \inf f(A)$$

Proof

Let, $x, y \in A$.

As,

$$\inf f(A) \leq f(x) \leq \sup f(A)$$

Thus,

$$f(x) - f(y) \leq \sup f(A) - \inf(A)$$

Since, $f(x) \leq \sup f(A), -f(y) \leq -\inf f(A)$.

As, x, y really didn't play a special role here, also have:

$f(y) - f(x) \leq \sup f(A) - \inf(A) \implies -(\sup f(A) - \inf f(B)) \leq f(x) - f(y)$

Hence, for all $x, y \in A$

$$|f(x) - f(y)| \leq \sup f(A) - \inf f(A)$$

For arbitrary closeness, we can utilize arbitrary closeness property.

So, there exists $x, y \in A$ such that:

$$0 \leq \sup f(A) - f(x) < \frac{\varepsilon}{2} \quad 0 \leq f(y) - \inf f(A) < \frac{\varepsilon}{2}$$

Thus,

$$0 \leq \sup f(A) - \inf f(A) - f(x) + f(y) < \varepsilon$$

Thus,

$$\sup f(A) - \inf f(A) - (f(x) - f(y)) < \varepsilon$$

Because, $-|f(x) - f(y)| \leq -(f(x) - f(y))$, we've:

$$\sup f(A) - \inf f(A) - |f(x) - f(y)| < \varepsilon$$

This shows arbitrary closeness which is an equivalent property to least upper bound property.

Thus, we're done.

□
